

Topic 2: Functional Relationships in Econometrics

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1 Conditional Expectation

Some conventions on notation for random variables

- We denote random variables by capital letters, i.e., X, Y, Z , etc.
- All random variables in the course will be real-valued.
- A realization or observed value of a random variable, will be denoted by lower-case letters, i.e., x, y, z .
- The p.d.f. of R.V. X is denoted by $f_X(x)$ or $f(x)$ when it is without confusion to what r.v. the p.d.f. refers to.
- The c.d.f. of R.V. X is denoted $Pr(X \leq x) := F_X(x)$ or $F(x)$ when it is without confusion to what r.v. the c.d.f. refers to.
- For the random vector X, Y the p.d.f. is denoted $f_{X,Y}(x, y)$ or $f(x, y)$. The CDF $Pr(X \leq x, Y \leq y) : F_{X,Y}(x, y)$ or $F(x, y)$.

• Conditional distributions :

$$f_{Y|X=x}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \quad \text{continuous case}$$

$$Pr(Y=y|X=x) = \frac{Pr(Y=y, X=x)}{Pr(X=x)} \quad \text{discrete case}$$

Remark : $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. f is linear if
 (i) $f(x+y) = f(x) + f(y)$ x, y both in $\mathbb{R}^m$
 (ii) $f(\alpha x) = \alpha f(x)$, where $\alpha \in \mathbb{R}$
 $x \in \mathbb{R}^m$

The conditional expectation:

$$E[Y|X_1, \dots, X_k]$$

is a very general formulation to define a functional relationship between variable Y in terms of explanatory variables $X_1, \dots, X_k$.

→ Substitute functional relationship:

Best linear predictor: $\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$

↓
 in the parameters because it has an intercept
 ↑
 this is called affine!

→ logit model : $\exp(\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k)$

↓
 used when Y is dichotomic/binary

Definition: Let (X, Y) be a random vector with p.d.f. $f(x, y)$.

Let $E_{\mathbf{y}}$:= expectation with respect to the marginal p.d.f. $f_{\mathbf{y}}$. Let $h(X, Y)$ be a random vector such that $E_y[h(X, Y)]$ exists. Then we define the conditional expectation by

$$E(h(X, Y)|X = x) = \int h(x, y)f(y|x)dy$$

where $f(y|x) = \frac{f(x, y)}{f_x(x)}$ is the conditional density function of Y given X .

- Notice $E(h(X, Y)|X = x)$ is only a function of x , say $\omega(x)$
- $E(h(X, Y)|X)$ then this means that we are calculating ω in the random argument X , i.e. $\omega(X) \rightarrow$ is a R.V., its distribution depends on the marginal p.d.f. $f_x(x)$

Theorem 2.1

Let (X, Y) be a r.v. such that $E[h(Y)] < \infty$ (exists and finite), then:

- (i) $E[g(X)h(Y)|X = x] = g(x)E[h(Y)|X = x]$
- (ii) $E[h_1(Y) + h_2(Y)|X = x] = E[h_1(Y)|X = x] + E[h_2(Y)|X = x]$
- (iii) (Law of iterated expectations): $E[h(Y)] = E(E(h(Y)|X))$

Note $E[h(y)|X]$ is taken w.r.t. $f(Y|X)$

outer expectation $E(E(h(Y)|X))$ is taken w.r.t. $F_x(x)$

Proof:

- (i) $E[g(X)h(Y)|X = x] = \int g(x)h(y)f(y|x) dy = g(x) \int h(y)f(y|x) dy$, note that $\int h(y)f(y|x) dy$ is $E[h(Y)|X = x]$

- (ii) homework

- (iii) $E(E(h(Y)|X)) = \int \overbrace{E[h(Y)|X = x]}^{\omega(x)} f_X(x) dx = \int \left[\int h(y)f(y|x) dy \right] f_X(x) dx = \int \int h(y) \underbrace{f(y|x)}_{\frac{f(x,y)}{f_X(x)}} dy \underbrace{f_X(x)}_{f_X(x)} dx = \int \int h(y) f(x,y) dy dx = \int \int h(y) f(x,y) dx dy = \int h(y) f_Y(y) dy = E[h(Y)]$
by definition $f_Y(y) = \int f(x,y) dx$.

$$E[h(Y)] = \int h(y) f_Y(y) dy$$

$\Rightarrow \int h(y) \left\{ \underbrace{\int f(x,y) dx}_{f_Y(y)} \right\} dy$

$$E[h(Y)|X=x] = \omega(x)$$

$$E[\omega(X)] = \int \omega(x) f_X(x) dx$$

The theorem (LIE) states:

$$\underline{E_Y[h(Y)]} = E_X[\omega(X)]$$

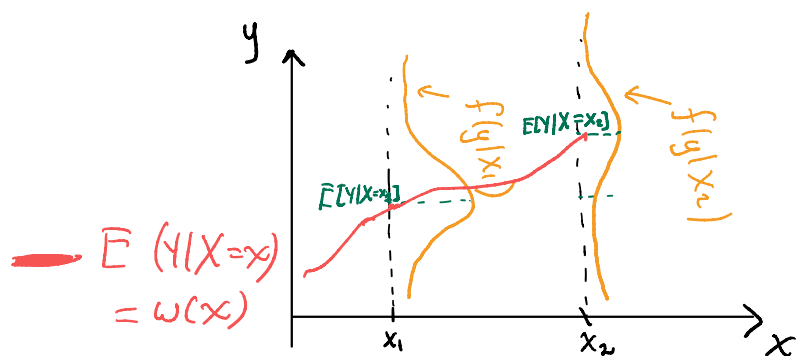
Unconditional exp.
of $h(Y)$

Iterated exp. by first
conditioning on X

Conditional Variance: variance of Y taken for the subpopulation $X = x$. The expectation of the mean squared-error has to be taken with respect to $E[Y|X = x]$ and with respect to the conditional p.d.f.

Definition 2.2 Let (X, Y) be a r.v. such that $E[Y^2] < \infty$. We define:

$$\text{Var}(Y|X = x) = \int [y - E[Y|X = x]]^2 f(y|x) dy = E[(Y - E[Y|X = x])^2 | X = x]$$



$$\mu_x = E[Y|X=x]$$

Theorem 2.2. Let (X, Y) be a r.v. such that $Var(Y) < \infty$, then:

$$(i) \quad Var(Y|X=x) = E[Y^2|X=x] - (E[Y|X=x])^2 = E[Y^2|X=x] - \mu_x^2$$

$$(ii) \quad Var(g(X)Y|X=x) = (g(x))^2 Var(Y|X=x)$$

$$(iii) \quad (\text{Law of Iterated Variance}): \quad Var(Y) = \underbrace{E[Var(Y|X)]}_{\sigma_x^2} + Var(\underbrace{E[Y|X]}_{\omega(X)})$$

Proof:

1. Homework

2. Homework

3. From the LIE (iii from Theorem 2.1.) $E[E[Y^2|X]] = E[Y^2]$.

$$\text{Also, } Var(\omega(X)) = E[(\omega(X))^2] - (E[\omega(X)])^2.$$

$$\text{Let } \omega(x) = E[Y|X=x], \text{ from (i) } Var(Y|X=x) = E[Y^2|X=x] - (\omega(X))^2$$

$$\Rightarrow (E[Var(Y|X)] = E[E[Y^2|X]] - E[(\omega(X))^2] = E[Y^2] - [Var(\omega(X)) + (E[\omega(X)])^2])$$

$$\therefore E[Var(Y|X)] = E[Y^2] - Var(\omega(X)) - [E[\omega(X)]]^2$$

$$\text{By LIE } E[\omega(X)] = E[E(Y|X)] = E[Y]$$

$$\therefore E[Var(Y|X)] = E[Y^2] - Var(\omega(X)) - [E[Y]]^2$$

$$\text{Notice that } E[Y^2] - [E[Y]]^2 \text{ is } Var(Y)$$

$$\Rightarrow E[Var(Y|X)] = Var(Y) - Var(\omega(X)) = Var(Y) - Var(E[Y|X])$$

$$\Rightarrow Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$$

Note: this property is related to the R^2 (R-squared) of the linear regression.

Covariance: measure of linear dependence between random variables.

$$Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$$

(unconditional covariance)

Definition 2.3. Let (X, Y, Z) be a r.v. such that $E[X] < \infty, E[Y] < \infty$. We define the conditional covariance of (X, Y) given $Z = z$ as

$$Cov(X, Y|Z = z) = E[(X - E[X|Z = z])(Y - E[Y|Z = z])|Z = z]$$

Note $E[X|Z = z]$ is taken w.r.t. $f(x|z) = \frac{f_{x,z}(x,z)}{f_z(z)} = \frac{\int f(x,y,z) dy}{\iint f(x,y,z) dx dy}$,
 $E[Y|Z = z]$ is taken w.r.t. $f(y|z) = \frac{f_{y,z}(y,z)}{f_z(z)} = \frac{\int f(x,y,z) dx}{\iint f(x,y,z) dx dy}$

The outer expectation is taken w.r.t. $f(x, y|z) = \frac{f(x,y,z)}{f_z(z)}$

Theorem 2.3.

- (i) $Cov(X, Y|Z) = Cov(Y, X|Z)$ (symmetry)
- (ii) $Cov(X, Y|Z) = E[XY|Z] - E[X|Z]E[Y|Z]$ (computing formula)
- (iii) $Cov(g(Z)X, Y|Z) = g(Z)Cov(X, Y|Z)$. Linearity.
- (iv) $Cov(X + Y, W|Z) = Cov(X, W|Z) + Cov(Y, W|Z)$ Linearity.
- (v) $Cov(X, X|Z) = Var(X|Z)$

Proof: (i), (iii), (iv), (v) are left as homework.

$$\begin{aligned} \text{(ii) } Cov(X, Y|Z) &= E([X - E(X|Z)][Y - E(Y|Z)]|Z) = E(XY - XE(Y|Z) - \\ &E[X|Z]Y + E[X|Z]E(Y|Z)|Z) = E(XY|Z) - E(XE(Y|Z)|Z) - E(E(X|Z)Y|Z) + \\ &E[E(X|Z)E(Y|Z)|Z] = E(XY|Z) - E(Y|Z)E(X|Z) - E(X|Z)E(Y|Z) + E(X|Z)E(Y|Z) = \\ &E(XY|Z) - E(X|Z)E(Y|Z). \end{aligned}$$

With the help of Theorem 2.3., we have the following result:

Theorem 2.4. $Var(X + Y|Z) = Var(X|Z) + Var(Y|Z) + 2Cov(X, Y|Z)$

Proof: Homework (start using (v) from Theorem 2.3, and then use the remaining properties)

2 Conditional Expectation as the Best Linear Predictor

Conditional Expectation as the Best Predictor

We will characterize $E(Y|X)$ as the best possible predictor of Y as a function of X .

Suppose we have a vector (X, Y) and we want to predict Y using a function of X , say $g(X)$.

Let $U(X, Y) = Y - g(X)$ be the prediction error. We wish to find a function $g(\cdot)$ such that $E[(U(X, Y))^2]$ is a minimum.

$E[(U(X, Y))^2]$ – "mean-squared prediction error"

Theorem 2.5. Let Y be a r.v. such that $Var(Y) < \infty$.

Then $\omega(x) = E[Y|X = x]$ is the best predictor of Y in the sense that it minimizes the mean-squared prediction error, that is

$$E[(Y - g(X))^2] \geq E[(Y - \omega(X))^2]$$

for any $g(X)$.

Proof: $U = Y - g(X) = Y - E[Y|X] + E[Y|X] - g(X)$.

Call $\varepsilon = Y - \omega(X)$, $D = \omega(X) - g(X)$, then $U = \varepsilon + D$ and $U^2 = \varepsilon^2 + 2\varepsilon D + D^2$

Recall by LIE, $E[U^2] = E(E[U^2|X])$ and using linearity of conditional expectation $E[U^2|X] = E[\varepsilon^2|X] + E[D^2|X] + 2E[\varepsilon D|X] = E[\varepsilon^2|X] + D^2 + 2DE[\varepsilon|X]$

For the moment, consider the following as fact: Fact: $E[\varepsilon|X] = 0$ for any X (will be proved later)

Therefore we have that: $E[U^2|X] = E[\varepsilon^2|X] + D^2$

Therefore, $E(U^2) = E(E(\varepsilon^2|X)) + E(D^2) = E(\varepsilon^2) + E(D^2) \geq E(\varepsilon^2)$

since $E(D^2) \geq 0$, and the proof is complete. ■

Suppose we take $g(X) = E(Y)$, i.e., we use as predictor the constant function equal to the expected value of Y .

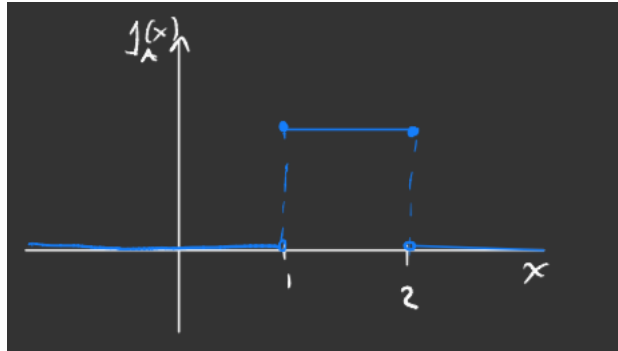
From Theorem 2.5. we got that

$$E[(Y - E(Y))^2] \geq E[(Y - E(Y|X))^2]$$

Notation: The indicator function of a set A is defined as follows:

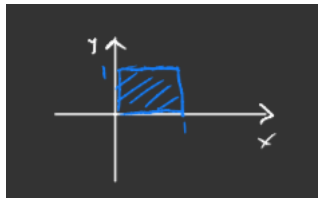
$$1_A(x) = \begin{cases} 0 & \text{if } x \notin A \\ 1 & \text{if } x \in A \end{cases}$$

Example $A = [1, 2]$



Example 2.1. Let X, Y be a r.v. with p.d.f.

$$f(x, y) = (x + y)1_{[0,1]}(x)1_{[0,1]}(y) dy$$



→ this r.v. has support (set with positive probability) on the unit square $[0, 1] \times [0, 1]$

In order to find the conditional p.d.f., we need to find f_x

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} (x+y) 1_{[0,1]}(x) 1_{[0,1]}(y) dy \\
&= \int_0^1 (x+y) 1_{[0,1]}(x) dy \\
&= 1_{[0,1]}(x) \int_0^1 (x+y) dy \\
&= 1_{[0,1]}(x) \left(xy + \frac{y^2}{2} \right)_0^1 \\
&= 1_{[0,1]}(x) (x + 1/2)
\end{aligned} \tag{1}$$

$$f(y|x) = \begin{cases} \frac{(x+y) 1_{[0,1]}(x) 1_{[0,1]}(y)}{(x+1/2) 1_{[0,1]}(x)} & \text{for } x \in [0, 1] \\ 0 & \text{if } x \notin [0, 1] \end{cases}$$

$$\Rightarrow f(y|x) = \frac{(x+y)}{(x+1/2)} 1_{[0,1]}(y)$$

$$\begin{aligned}
E[Y|X=x] &= \int_{-\infty}^{\infty} y \frac{(x+y)}{(x+1/2)} 1_{[0,1]}(y) dy \\
&= \int_0^1 y \frac{(x+y)}{(x+1/2)} dy \\
&= \frac{1}{(x+1/2)} \left(x \int_0^1 dy + \int_0^1 y^2 dy \right) \\
&= \frac{1}{(x+1/2)} \left(x \left(\frac{y^2}{2} \right)_0^1 + \left(\frac{y^3}{3} \right)_0^1 \right) \\
&= \frac{x(1/2) + 1/3}{x+1/2} \\
&= \frac{\frac{3x+2}{6}}{\frac{6x+3}{6}} \\
&= \frac{3x+2}{6x+3}
\end{aligned} \tag{2}$$

The prediction error, as we have seen, is $\varepsilon = Y - E(Y|X)$
Theorem 2.6. Let $\varepsilon = Y - E(Y|X)$ be the part of Y not explained by $E(Y|X)$, then:

- (i) $E(\varepsilon|X) = 0$
- (ii) $E(\varepsilon) = 0$
- (iii) $Var(\varepsilon|X) = Var(Y|X)$ and $Var(\varepsilon) = E[Var(Y|X)]$
- (iv) $Var(\varepsilon) \leq Var(Y)$ always
- (v) Let $g(X)$ be any function of X such that $Var(g(x)) < \infty$
Then $Cov(\varepsilon, g(x)) = 0$.

Proof:

- (i) $E(\varepsilon|X) = E(Y - E(Y|X)|X) = E(Y|X) - E(Y|X) = 0$
- (ii) Straightforward from (i) and LIE. (Homework, verify this!)
- (iii) Homework (Application of Law of Iterated Variance)
- (iv) Homework
- (v) $Cov(\varepsilon, g(x)) = E(\varepsilon g(x)) - E(\varepsilon)E(g(x)) = E(\varepsilon g(x))$
Homework: show that $E(\varepsilon g(x)) = 0$ (LIE), $\therefore$ the assertion is true.

3 The Best Linear Predictor

The Best Linear Predictor

The idea is the following:

Find $g(x) = a + bx$ a linear function of x such that $g(\cdot)$ minimizes

$$E[(Y - g(X))^2]$$

Theorem 2.7. Let (X, Y) be a r.v. such that $0 < \text{Var}(X) < \infty$, $\text{Var}(Y) < \infty$. Then the values of a and b that minimize $E[(Y - a - bX)^2]$ are:

$$\alpha = E[Y] - \beta E[X]$$

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

Proof: Let $MSE(a, b) = E[(Y - a - bX)^2]$
 If α and β are optimal, in the sense of minimizing the MSE, then they must satisfy the First Order Conditions:

$$\frac{\partial MSE}{\partial a}(\alpha, \beta) = 0$$

$$\frac{\partial MSE}{\partial b}(\alpha, \beta) = 0$$

Under some regularity conditions on the joint distribution $f(x, y)$ (technical), we can exchange the differentiation operator with the expectation operator:

$$\frac{\partial MSE}{\partial a}(a, b) = E\left[\frac{\partial}{\partial a}(Y - a - bX)^2\right] = -2E[Y - a - bX]$$

$$\frac{\partial MSE}{\partial b}(a, b) = E\left[\frac{\partial}{\partial b}(Y - a - bX)^2\right] = -2E[(Y - a - bX)X]$$

Hence, the FOCs are:

$$E[Y - \alpha - \beta X] = 0$$

$$E[(Y - \alpha - \beta X)X] = 0$$

Let $U^* = Y - \alpha - \beta X$, then we can rewrite the above as:

$$E[U^*] = 0 \quad (1)$$

$$E[U^*X] = 0 \quad (2)$$

(1) is equivalent to $E[Y] - \alpha - \beta E[X] = 0 \Rightarrow \alpha = E[Y] - \beta E[X]$

Now: $Cov(U^*, X) = E[U^*X] - E[U^*]E[X] = 0, \Rightarrow Cov(U^*, X) = 0$

$Cov(U^*, X) = 0 = Cov(Y - \alpha - \beta X, X) = Cov(Y, X) - \beta Cov(X, X) = Cov(Y, X) - \beta Var(X) \Rightarrow \beta = \frac{Cov(Y, X)}{Var(X)}, \therefore$ the theorem is true.

Note: We would need to check the second-order conditions to see that (α, β) are a global minimum (convexity of MSE(a,b))

The line $\alpha + \beta X$ is called the Best Linear Predictor of Y as a function of X.
It is denoted as:

$$E^*(Y|x) = \alpha + \beta X$$

→ This is not an expectation operator, but it is only notation.

alternative notation: $L^*(Y|X) = \alpha + \beta X$

Example 2.2. Let (X, Y) be r.v. with p.d.f. given by

$$f(x, y) = (x + y)1_{[0,1]}(x)1_{[0,1]}(y)$$

(same as in ex. 2.1.) I will leave as homework (will appear in a future problem set) to show that

$$E[X^m Y^n] = \frac{(m+1)(n+2) + (m+2)(n+1)}{(m+1)(m+2)(n+1)(n+2)}$$

$$E(X) = E(Y) = \frac{7}{12};$$

$$E(X^2) = \frac{5}{12};$$

$$E(XY) = \frac{1}{3};$$

$$Var(X) = E(X^2) - (E(X))^2 = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144}$$

$$Cov(X, Y) = E(XY) - E[X]E[Y] = \frac{1}{3} - \left(\frac{7}{12}\right)^2 = -\frac{1}{144}$$

$$\beta = \frac{Cov(X, Y)}{Var(X)} = \frac{-1/144}{11/144} = -1/144 \approx -0.0909$$

$$\alpha = E(Y) - \beta E(X) = \frac{7}{12} - \left(-\frac{1}{11}\right)\left(\frac{7}{12}\right) = \frac{7}{12} + \frac{7}{132} = \frac{84}{132} = \frac{7}{11}$$

∴ The BLP is

$$E^*(Y|X) = \frac{7}{11} - \frac{1}{11}x$$

Using the BLP, the effect of X on Y would be:

$$\frac{\partial E^*(Y|X)}{\partial X} = -\frac{1}{11}$$

which estimates the true effect $\frac{\partial E(Y|X)}{\partial X}$
BLP is a linear approximation to $E(Y|X)$

Recall "Prediction error by the BLP" $= U^* = Y - E^*(Y|X) = Y - \alpha - \beta X$, where $\alpha = E(Y) - \beta E(X)$, $\beta = \frac{Cov(X,Y)}{Var(X)}$
Theorem 2.8. Let (X,Y) be a r.v. such that $Var(X) < \infty, Var(Y) < \infty$. Let $U^* = Y - E^*(Y|X)$, then

(i) $E[U^*] = 0$

(ii) $Cov(X, U^*) = 0$

(iii) $Var(U^*) = Var(Y) - \beta^2 Var(X)$

Proof: (i) and (ii) follow from the FOC's from Theorem 2.7.

For (iii)

$$Var(U^*) = Var(Y - \alpha - \beta X) = Var(Y) + \beta^2 Var(X) - 2\beta Cov(Y, X)$$

$$\beta = \frac{Cov(X,Y)}{Var(X)} \Rightarrow \beta Var(X) = Cov(X, Y) \Rightarrow Var(U^*) = Var(Y) + \beta^2 Var(X) - 2\beta^2 Var(X) = Var(Y) - \beta^2 Var(X)$$

■

Comments:

Although $E[U^*] = 0$, $E[U^*|x]$ is not necessarily zero

Homework: $E[U^*|X] = E[Y|X] - E^*(Y|X)$

Note : $E[U^*|X] < 0$

> 0

$= 0$

- (ii) Linear correlation between X and U^* is zero $\therefore$ no improvement of the BLP can be obtained from using another linear function of X
- (iii) gives the reduction in prediction variance due to the inclusion of X in a linear manner.

Theorem 2.9. Let (X, Y) be a r.v. such that $Var(X) < \infty, Var(Y) < \infty$. If $E(Y|X) = m + bX$, then $E(Y|X) = E^*(Y|X)$

Proof: Suppose $E[Y|X] = m + bX$, we will show that $\alpha = m, \beta = b$.

Notice that:

$$\begin{aligned}
 E(Y) &= E(E(Y|X)) = E(m + bX) = m + bE(X) \\
 E(XY) &= E(E(XY|X)) \\
 &= E(XE(Y|X)) \\
 &= E(X(m + bX)) \\
 &= E(mX + bX^2) \\
 &= mE(X) + bE(X^2) \\
 Cov(X, Y) &= E(XY) - E(X)E(Y) \\
 &= mE(X) + bE(X^2) - E(X)(m + bE(X)) \\
 &= mE(X) + bE(X^2) - mE(X) - b(E(X))^2 \\
 &= b(E(X^2) - [E(X)]^2) \\
 &= bVar(X)
 \end{aligned}$$

The previous implies that $\beta = b$ and also $\alpha = m$. We will extend the definition of the BLP to several explanatory variables. To do that we need the notions of the variance-covariance matrix of a random vector, and of the Covariance between a random vector and a random variable.

Definition 2.4: Let $\mathbf{X} = (X_1, X_2, \dots, X_k)^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}$ be k-dimensional vector

with finite variances. We define the variance-covariance matrix by

$$V(\mathbf{X}) = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots & Cov(X_1, X_k) \\ Cov(X_2, X_1) & Var(X_2) & \dots & Cov(X_2, X_k) \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & Var(X_k) \end{bmatrix}$$

$V(\mathbf{X})$ is k×k matrix in which:

- entry (i,i), i.e., entries in the diagonal, correspond to $Var(X_i)$
- entries (i, j), with $i \neq j$, i.e., entries off-diagonal, corresponds to $Cov(X_i, X_j)$
- $V(\mathbf{X})$ is symmetric because $Cov(X_i, X_j) = Cov(X_j, X_i)$
 $[V(\mathbf{X})]^T = V(\mathbf{X})$

If Y is a random variable we define

$$Cov(\mathbf{X}, Y) = \begin{bmatrix} Cov(X_1, Y) \\ Cov(X_2, Y) \\ \vdots \\ Cov(X_k, Y) \end{bmatrix}$$

With this notation, we can now generalize the BLP to several explanatory variables.

The vector of coefficients (α, β) of the BLP with a single explanatory variable correspond to: α = intercept, β = slope. We are going to denote the vector of coefficients of the general BLP as $\beta = (\beta_0, \beta_1, \dots, \beta_k)^T$ where β_0 is the intercept and $\beta_{\mathbf{p}} = (\beta_1, \dots, \beta_k)^T$ is the vector of slopes.

Theorem 2.10 Let $\mathbf{X}$ be a r.v. with finite variables and $\det[V(\mathbf{X})] > 0$. Then the values of $(c_0, c_1, \dots, c_k)$ that minimize $E[(Y - c_0 - c_1x_1 - c_2x_2 - \dots - c_kx_k)^2]$ are given by

$$\begin{aligned}\beta_0 &= E[Y] - \beta_1 E[X_1] - \dots - \beta_k E[X_k] \\ \beta_p &= [V(X)]^{-1} Cov(\mathbf{X}, Y)\end{aligned}$$

Proof: Let us denote $MSE(C_0, C_1, \dots, C_k) = E[(Y - C_0 - C_1X_1 - \dots - C_kX_k)^2]$. Exchange the differentiation operator with the expectation operator, the partial derivatives are:

$$\begin{aligned}\frac{\partial}{\partial C_0} MSE(C_0, C_1, \dots, C_k) &= E[-2(Y - C_0 - C_1X_1 - \dots - C_kX_k)] \\ \text{for } i = 1, \dots, k \quad \frac{\partial}{\partial C_i} MSE(C_0, C_1, \dots, C_k) &= E[-2(Y - C_0 - C_1X_1 - \dots - C_kX_k)X_i]\end{aligned}$$

In order to write the FOC's let $U^* = Y - \beta_0 - \beta_1 X_1 - \dots - \beta_k X_k$ be the prediction error of the BLP, then the FOC's become:

$$\begin{aligned} \frac{\partial}{\partial C_0} MSE(\beta) &= E[-2U^*] = 0 = E[U^*] = 0 \\ \text{for } i = 1, \dots, k \quad \frac{\partial}{\partial C_i} MSE(\beta) &= E[-2X_i U^*] = 0 = E[X_i U^*] = 0 \end{aligned}$$

As with the proof of Theorem 2.7, we have that for each $i = 1, \dots, k$

$$Cov(U^*, X_i) = E[U^* X_i] - E[U^*]E[X_i] = 0$$

but $Cov(U^*, X_i) = Cov(Y - \beta_0 - \beta_1 X_1 - \dots - \beta_k X_k, X_i); i = 1, \dots, k$

All of the previous observations lead to the following system of equations

$$\begin{aligned}
& \beta_0 + \beta_1 E(X_1) + \dots + \beta_k E(X_k) = E(Y) \\
i = 1 & \beta_1 Var(X_1) + \beta_2 Cov(X_1, X_2) + \dots + \beta_k Cov(X_1, X_k) = Cov(X_1, Y) \\
i = 2 & \beta_1 Cov(X_2, X_1) + \beta_2 Var(X_2) + \dots + \beta_k Cov(X_2, X_k) = Cov(X_2, Y) \\
& \vdots \\
i = k & \beta_1 Cov(X_k, X_1) + \beta_2 Cov(X_k, X_2) + \dots + \beta_k Var(X_k) = Cov(X_k, Y)
\end{aligned}$$

writing the last equations in matrix form yields:

$$\begin{bmatrix} Var(X_1) & Cov(X, X_2) & \dots & Cov(X_1, X_k) \\ Cov(X_2, X_1) & Var(X_2) & \dots & Cov(X_2, X_k) \\ \vdots & \ddots & \ddots & \vdots \\ Cov(X_k, X_1) & Cov(X_k, X_2) & \dots & Var(X_k) \end{bmatrix} \times \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} = \begin{bmatrix} Cov(X_1, Y) \\ Cov(X_2, Y) \\ \vdots \\ Cov(X_k, Y) \end{bmatrix}$$

OR

$$V(\mathbf{X})\beta_{\mathbf{p}} = Cov(\mathbf{X}, Y) \Rightarrow \beta_{\mathbf{p}} = [V(\mathbf{X})]^{-1}Cov(\mathbf{X}, Y)$$

■

Example 2.3. Once again let (X, Y) be the r.v. with p.d.f. $f(x, y) = (x + y)1_{[0,1]}(x)1_{[0,1]}(y)$. We build a better BLP by constituting a relationship between Y and X, X^2 that is:

$$E^*(Y|X, X^2) = \beta_0 + \beta_1 X + \beta_2 X^2$$

In this case $\mathbf{X} = (X, X^2)^T$. We need to find $V(\mathbf{X})$, and $Cov(\mathbf{X}, \mathbf{Y})$ which consists of $Cov(X, Y)$ and $Cov(X^2, Y)$,

$$V(\mathbf{X}) = \begin{bmatrix} Var(X) & Cov(X, X^2) \\ Cov(X, X^2) & Var(X^2) \end{bmatrix}$$

Recall $E(X^m Y^n) = \frac{(m+1)(n+1)+(m+2)(n+1)}{(m+1)(m+2)(n+1)(n+2)}$

$$\begin{aligned}
Cov(X, X^2) &= E(X^3) - E(X)E(X^2) = \frac{13}{40} - \frac{7}{12} \frac{5}{12} \\
Var(X^2) &= E(X^4) - (E(X^2))^2 = \frac{16}{60} - \left(\frac{5}{12}\right)^2 \\
Cov(X^2, Y) &= E(X^2, Y) - E(X^2)E(Y) = \frac{17}{72} - \frac{5}{12} \frac{7}{12}
\end{aligned}$$

We obtain that

$$E^*(Y|X, X^2) = 0.6568 - 0.19607X + 0.098039X^2$$

→ The effect of X on Y is:

$$\frac{\partial E^*(Y|X, X^2)}{\partial X} = \beta_1 X + 2\beta_2 X = -0.19607 + 2(0.098039)X$$

This approximation is better than the value given $\frac{\partial E^*(Y|X)}{\partial X}$.